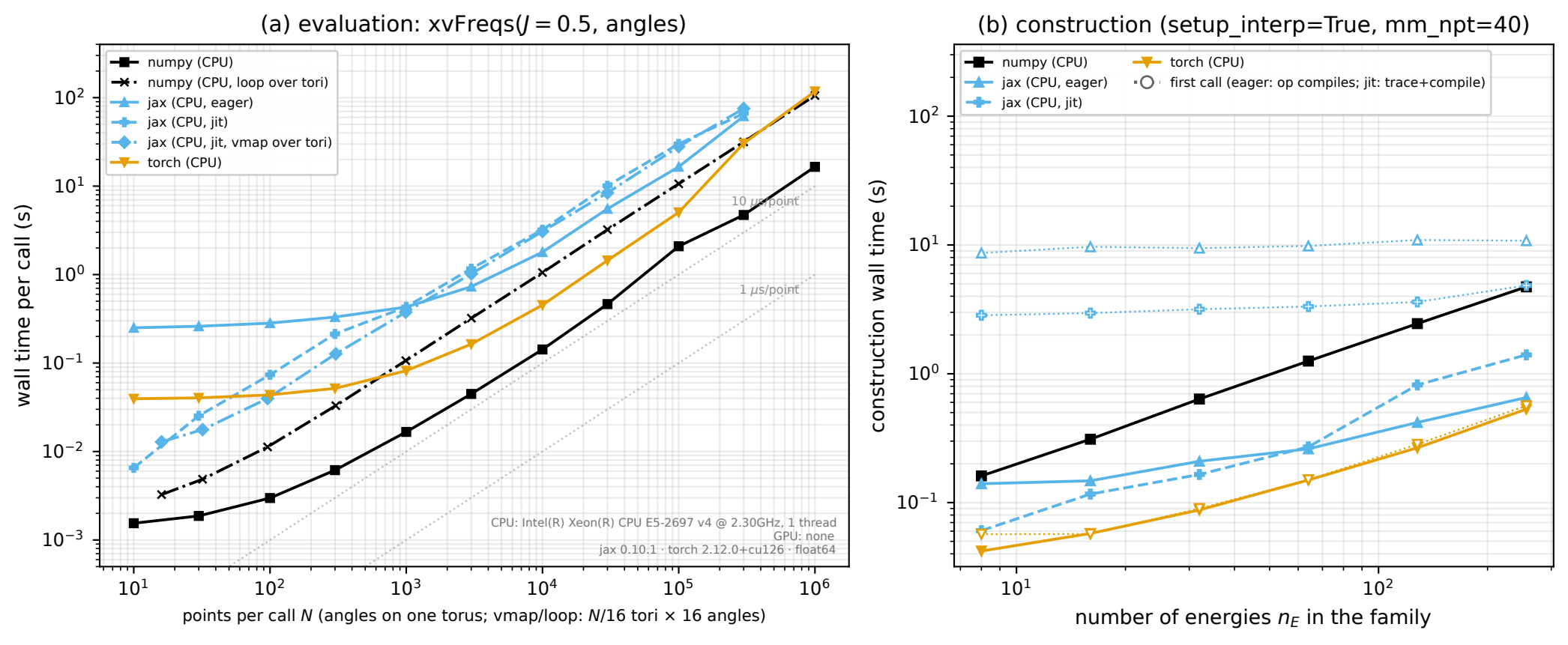
(a) evaluation: xvFreqs($J = 0.5$, angles)
numpy (CPU)
numpy (CPU, loop over tori)
jax (CPU, eager)
jax (CPU, jit)
jax (CPU, jit, vmap over tori)
torch (CPU)
10 μs/point
1 μs/point
CPU: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz, 1 thread
GPU: none
jax 0.10.1 · torch 2.12.0+cu126 · float64
wall time per call (s)
points per call N (angles on one torus; vmap/loop: N/16 tori × 16 angles)
(b) construction (setup_interp=True, mm_npt=40)
numpy (CPU)
jax (CPU, eager)
jax (CPU, jit)
torch (CPU)
first call (eager: op compiles; jit: trace+compile)
construction wall time (s)
number of energies n_E in the family